



EDUCATION & SCHOLASTIC ACHIEVEMENTS			
Program	Institute	Percentage/ CGPA	Graduation
Dual degree in Mechanical Engineering	Indian Institute of Technology Madras		2021
Class XII (MH. SSC)	S.S. High School & Jr. College, Mumbai		2016
Class X (C.B.S.E.)	D.A.V. Public School, New Panvel		2014

- **Patent¹:** 'An electromagnetic system and device utilizing gravitational potential to condense atmospheric water' **IN201941026443**
- **Conference Publication:** 16th ACFM "Exploring Physics Informed Neural Networks for Compressible flow predictions" in Dec, 2019
- Scholar stipend from **Singapore Ministry of Manpower** to pursue research at **Temasek Lab, NUS** (country's #1 academic defense lab)

PROFESSIONAL EXPERIENCE	
Reliance Industries Engineering Intern May – July 2020	<i>Proposed immediate measures to improve efficiency of C3R Compressor - Steam Turbine couple at NMD power plant</i> • Identified 5.5% overall efficiency gap leading to raised C3R flow through the cycle & 3% higher operating cost • Bridged 40% of the gap prognostically by suggesting immediate governor re-calibration & turbine leak check • Advised impeller replacement in the long term by studying maintenance cycles of Compressor's fouled blades • Reduced 60% inlet-side bearing overheating temporarily by proposing 12% more lubricant flow provision
National Univ. of Singapore (QS #11) UG Research Intern May – July 2019	<i>Built Physics Informed Neural Networks (PINNs) for 3 applied aerodynamic problems of defense aircrafts at T-Lab</i> • Obtained 94% prediction accuracy with 1% training data in 'Physics learnt'² Neural Networks for 1D flow shocks • Achieved 200% better performance vs classic ANN through tuning 5 hyperparameters in scarce CFD datasets • Implemented a 96% accurate time marching RNN to model real-time motion of 2D airfoil wings during flight
Nihon Technology Blockchain Developer May – June 2018	<i>Pioneered the development of a blockchain powered, reward-based e-learning platform (LearnChain) on Hyperledger</i> • Created a smart contract based digital currency for a Rs. 10L+ ICO with 100% token exchange rate control • Led the Product delivery team to develop an e-wallet interface (10k user capacity) to trade cryptos via PayPal
Kore Health Finland Product Intern May – June 2020	<i>Drove GoKore UX (app to speak out one's overwhelming inner voices) by spearheading v0.0.1 development & launch</i> • Boosted user engagement by 35% by introducing 14 fiction character voices relating to 9 most common emotions • Increased user intimacy³ by designing 15+ emotional cues for 3 strongest negative states – fear, sadness & anger

KEY PROJECTS	
Ejector Cooler Dual Degree Thesis	<i>Simulating Ejector refrigeration systems to reuse 65+% of waste heat from solar power generation for cold storage</i> • Established correlation between suction flow rate & solar source temperature for dry air from preliminary tests
Sandwich Maker ME3302 Project Subway Feb – May 2019	<i>Redesigned the Subway's sandwich making process by automating the current manual sequence as a Flexible task</i> • Raised the throughput by 120%, reduced occupied area by 50%+ & cut laborious manual intervention to 20% • Rectified 4 bottlenecks in sandwich making by modelling & optimizing sub-processes in Siemens Tecnomatix • Devised this new process in a machine, yielding a Rs. 2.1L MVP cost with lesser than 60-day breakeven⁴ period
ICE Phasing Model Guide: Dr. A. Ramesh Sept – Dec 2018	<i>Simulated a novel model for angular phasing from Variable Reluctance type speed sensor for Digital Twin of IC Engine</i> • Reviewed literature on Bosch's VR speed sensing & derived the governing mathematical equations from scratch • Reproduced the phasing plot in MATLAB and yielded over 90% accuracy compared against experimental lab data
Solar Desalination Guide: Dr. K.S. Reddy Sept – Nov 2018	<i>Proposed a 400 Million liter per day Solar Seawater Desalination plant for Chennai working (33% of the demand)</i> • Designed a 10% more efficient 600 MW Sodium salt thermal cycle based, off-grid, concentrated 3.3km² Solar farm • Coupled a 400MLD RO based desalination system for city wide pumping, filtration, purification and pressurization

INTERNATIONAL COMPETITIONS	
IXL Olympics COVID-19 Policy	• Working with Colombia DC mayor to draft policies to create 200,000+ jobs in Latin American cities like Bogota • Proposing 6 actionable on MSME PPPs, safe recycling & building trust for the worst hit peri-urbans of COVID-19 • Top 6 university team worldwide from 1500+ yearly applications in the finals assisting a whitepaper publication
Miscellaneous	• Top 20 team worldwide in TFF⁵ Challenge '19 with sponsored invitation to Kuala Lumpur, MSY for final round • Top 15 team nationwide in Finnovatica '18 (Fullerton) & invitation to present to the senior team at Mumbai • Secured 4th regional team rank (Stellar Participant) in NASA Space Apps challenge 2017 conducted at VIT

SOCIAL VOLUNTEERING & ARTS		
Social Technology Innovations	Moisture Condenser ⁶	• Invented a novel gadget to provide urgent potable water in off-grid semi-arids (5mL/min rate) • Reduced 85% raw material price & 60% weight through redesigning using recyclable ABS
	APM Mattress	• Designed Rs. 3k adaptive air-mattress to prevent 95% skin ulcers in secondary hospital settings
	Learn AiR	• Qualified to HULT Prize (UN) pitching round by proposing (Rs. 6k) Edu-tab for 1 million students
Writing & Oratory		• Singlehandedly posted 110+ motivational video daily streak⁷ series on Insta channel over a period of 119 days • Featured as the top writer out of 50+ entries in Writing Club's Micro-fiction for Literati Magazine, March '19 Ed. • Achieved 160-day Spanish streak (Duolingo) in lockdown & now can communicate with 42+% of the world
Extensive Reading		• Read 60+ non-fiction works over 25 months from philosophy, spirituality, biographies, management practice
Finance		• Delegated as top 5% applicant for Transforming Investment Banking Conference (NatWest Markets), Park Hyatt • Mentored a social venture of 5 local school girls under E-Cell E-21 initiative, standing the team 2nd in the finals

POSITIONS OF RESPONSIBILITY		
E-Cell Events, Networking and Outreach	E-Summit '18	• Achieved 160%+ participation in yearly talk shows by rebranding to 'Inspirit Lecture Series' • Spearheaded a team of 8 associates to confirm 20+ speakers & organize the 3-day logistics
	Events	• Boosted E-Cell brand by organizing Hack2Innovate⁸ with NITI Aayog - 50+ participant colleges
Placement Team Dpt. Placement Coord.		• Streamlined 300+ student interviews & coordinated with HR officials from 20+ firms in first 5 days of Placements • Held direct responsibility for 10 seniors⁹ & handled logistics of the biggest dept. with 170+ record offers

COURSEWORK & SKILLS			
Key Courses	Automation in Manufacturing; Renewable Energy Technologies; Applied Thermal Engineering; Materials & Design		
Software	MATLAB; Fusion 360; SolidWorks; Siemens Tecnomatix	Languages	C, C++, Python (TF, Pandas, NP), R, Java

